

MODULE 13 · DOMAIN 3 - CLOUD TECHNOLOGY AND SERVICES

Storage

Three shapes of storage, seven S3 classes, and knowing which question is being asked.

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Object, block or file. **Almost every storage question is settled by which of the three the workload needs.**

The three shapes

SHAPE	LOOKS LIKE	AWS SERVICE
Object	Items fetched by name over HTTP	Amazon S3
Block	A disk attached to one machine	Amazon EBS, instance store
File	A shared folder many machines mount	Amazon EFS, Amazon FSx

ON THE EXAM

If an application must mount a filesystem without being changed, S3 is eliminated no matter how attractive it looks.

Amazon S3

- ◆ **Objects live in buckets.** Bucket names are globally unique; the bucket itself lives in one Region.
- ◆ **Eleven nines of durability**, achieved by storing redundantly across at least three Availability Zones.
- ◆ Storage is effectively unlimited. A single object can be up to five terabytes.
- ◆ Versioning keeps previous copies. Object Lock makes objects immutable for compliance.
- ◆ It is not a filesystem. You cannot mount it and applications cannot seek within an object.

ON THE EXAM

Eleven nines is durability, meaning the object will not be lost. It is not a statement about being able to reach it.

Durability or availability?

DURABILITY

Will the data still exist?

- Eleven nines for S3 standard classes
- About surviving hardware failure and loss
- Redundant copies across Availability Zones
- Almost identical across S3 classes

AVAILABILITY

Can I reach it right now?

- 99.99% for S3 Standard, lower for others
- About the service responding to a request
- Affected by how many AZs hold the data
- This is what actually differs between classes

THE TIEBREAKER

Durability barely changes between S3 classes. Availability does. That is what One Zone classes trade away.

The S3 storage classes

CLASS	USE IT WHEN
S3 Standard	Frequent access, milliseconds, no retrieval fee
S3 Intelligent-Tiering	Access pattern is unknown or changing. AWS moves it for you
S3 Standard-IA	Infrequent access, still needed in milliseconds
S3 One Zone-IA	Infrequent, and you could recreate it if the AZ was lost
S3 Glacier Instant Retrieval	Archive that must still come back in milliseconds
S3 Glacier Flexible Retrieval	Archive; minutes to hours is acceptable
S3 Glacier Deep Archive	Lowest cost; twelve to forty-eight hours is acceptable

ON THE EXAM

Read the scenario for two facts: how often the data is read, and how fast it must come back. Those pick the class.

Intelligent-Tiering or a lifecycle policy?

S3 INTELLIGENT-TIERING

AWS decides, by observing

- Moves objects based on actual access patterns
- Small monitoring charge per object
- No retrieval fees between its tiers
- For when you do not know the pattern

LIFECYCLE POLICY

You decide, by age

- Transitions objects after a number of days
- No monitoring charge
- You must know the pattern in advance
- Also used to expire and delete objects

THE TIEBREAKER

Unknown or unpredictable access means Intelligent-Tiering. A known age-based rule means a lifecycle policy.

Amazon EBS

- ◆ **A volume attaches to one EC2 instance** and must be in the same Availability Zone as it.
- ◆ It persists independently of the instance. Terminating the instance need not destroy the volume.
- ◆ **SSD types** suit databases and boot volumes. **HDD types** suit large sequential throughput.
- ◆ Volumes can be encrypted with KMS, and snapshots inherit that encryption.
- ◆ A stopped instance stops compute charges. Its EBS volume keeps billing.

ON THE EXAM

Same Availability Zone as the instance is a hard constraint. EBS is not shared storage across a Region.

EBS or instance store?

AMAZON EBS

Survives the instance

- Network-attached, persists independently
- Can be detached and reattached
- Snapshots to Amazon S3
- The default for anything you care about

INSTANCE STORE

Dies with the instance

- Physically attached to the host, very fast
- Data lost when the instance stops or terminates
- Cannot be detached or snapshotted
- For caches, buffers and scratch data

THE TIEBREAKER

Data must survive a stop or restart means EBS. Temporary scratch data that can be regenerated means instance store.

EBS snapshots

- ◆ **A snapshot is a point-in-time backup** of a volume, stored in Amazon S3.
- ◆ Snapshots are incremental. Only blocks changed since the last one are stored.
- ◆ **They can be copied to another Region**, which is how EBS data crosses Region boundaries.
- ◆ A new volume created from a snapshot can be placed in a different Availability Zone.
- ◆ This is the standard way to move a volume between AZs, since volumes themselves cannot move.

ON THE EXAM

Moving an EBS volume to another AZ or Region always means taking a snapshot and restoring from it.

Amazon EFS or Amazon FSx?

AMAZON EFS

Linux and NFS

- Mounted by many Linux instances at once
- Elastic; grows and shrinks automatically
- Spans multiple Availability Zones
- The default shared filesystem for Linux

AMAZON FSX

Windows, Lustre and others

- FSx for Windows File Server speaks SMB
- FSx for Lustre for high performance computing
- Native Windows features such as Active Directory
- Chosen when a specific protocol is required

THE TIEBREAKER

Linux and NFS means EFS. Windows and SMB means FSx for Windows File Server. The protocol names the service.

AWS Storage Gateway

- ◆ It gives on-premises systems access to AWS storage using protocols they already speak.
- ◆ File Gateway presents S3 as an NFS or SMB share.
- ◆ Volume Gateway presents block storage over iSCSI, backed by S3.
- ◆ Tape Gateway replaces a physical tape library, backed by S3 and Glacier classes.
- ◆ The signal is always an existing on-premises system that cannot be changed.

ON THE EXAM

Tape Gateway is the giveaway when a scenario mentions replacing physical tape backups without changing the backup software.

AWS Backup and Elastic Disaster Recovery

- ◆ **AWS Backup centralises backup** across EBS, RDS, DynamoDB, EFS, FSx and more, from one place.
- ◆ Backup plans define schedule and retention, and can be applied across an organisation.
- ◆ **AWS Elastic Disaster Recovery** continuously replicates servers into AWS for rapid recovery.
- ◆ It is the answer when a scenario names a recovery time objective for a whole server estate.
- ◆ A backup is a copy you restore from. Disaster recovery is a standby you fail over to. Different questions.

ON THE EXAM

Centralised backup policy across many services means AWS Backup. Failing an estate over to AWS means Elastic Disaster Recovery.

A note on the word Glacier

- ◆ **The S3 Glacier storage classes are fully available** and are the recommended way to archive.
- ◆ **Amazon Glacier, the original standalone vault-based service, stopped accepting new customers on 15 December 2025.**
- ◆ Existing vault data stays accessible indefinitely and no migration is required.
- ◆ They are different things with different APIs, and the exam means the storage classes.
- ◆ If a question mentions vaults and archives rather than buckets and objects, it is describing the old service.

ON THE EXAM

Vaults means the legacy standalone service. Buckets and storage classes means what the exam is testing.

Reverse lookup

IF THE SCENARIO SAYS	THE ANSWER IS
Many Linux instances need the same files	Amazon EFS
Windows applications need an SMB share	Amazon FSx for Windows File Server
A single instance needs a persistent disk	Amazon EBS
Store any amount of data, retrieved over HTTP	Amazon S3
Archive, retrieval within twelve hours acceptable	S3 Glacier Deep Archive
Access pattern is unpredictable	S3 Intelligent-Tiering
Replace an on-premises tape library	AWS Storage Gateway, Tape Gateway
Central backup policy across many services	AWS Backup

ON THE EXAM

Eight phrasings that cover most storage questions. Protocol words and retrieval times do most of the deciding.

A company stores legal case files in Amazon S3. Regulation requires them to be kept for ten years. In practice they are read a handful of times in the first ninety days and then almost never, though a request can arrive at any time and must be satisfied within a day.

The company wants the lowest storage cost that meets this.

Which lifecycle approach should the company use?

- ☐ A Keep everything in S3 Standard for ten years
- ☐ B Transition to S3 Glacier Deep Archive after ninety days
- ☐ C Transition to S3 One Zone-IA after ninety days
- ☐ D Transition to S3 Glacier Instant Retrieval after ninety days

◆ PAUSE AND DECIDE

Which lifecycle approach should the company use?

- A

 Keep everything in S3 Standard for ten years
Meets every access requirement and is by far the most expensive. Fails the stated goal.
- B

 Transition to S3 Glacier Deep Archive after ninety days
Cheapest storage, and standard retrieval completes within twelve hours, which is inside the one-day requirement.
- C

 Transition to S3 One Zone-IA after ninety days
Cheaper than Standard but far dearer than Glacier classes, and legal records should not sit in one AZ.
- D

 Transition to S3 Glacier Instant Retrieval after ninety days
Correct access pattern, but you pay for millisecond retrieval the company does not need.

THE SIGNAL

Within a day is a generous window. Deep Archive's twelve-hour standard retrieval fits inside it comfortably.

A company runs a content management application on twelve EC2 instances across three Availability Zones. All twelve must read and write the same set of uploaded documents, and the application expects a mounted filesystem path. The instances run Linux.

The application cannot be modified.

Which storage service should the company use?

- | | |
|---------------------------------------|---|
| ☐ A Amazon S3 | ☐ B Amazon EBS |
| ☐ C Amazon EFS | ☐ D Instance store |

◆ PAUSE AND DECIDE

Which storage service should the company use?

A Amazon S3

The trap. Cheap and durable, but it is not mountable, so the application would have to be rewritten.

B Amazon EBS

Attaches to one instance in one Availability Zone. It cannot be the shared store for twelve instances across three zones.

C Amazon EFS

A mountable NFS filesystem, shared by many Linux instances, spanning Availability Zones.

D Instance store

Local to one host and lost when the instance stops. Neither shared nor persistent.

THE SIGNAL

Expects a mounted filesystem path plus cannot be modified together eliminate S3.
That pairing is deliberate.

A data science team writes large temporary files while a job runs. The files are regenerated on every run, are never needed after the job finishes, and the team wants the fastest possible disk throughput.

Cost matters, and they are willing to lose the data at any time.

Which storage should they use for these temporary files?

- | | |
|---|---|
| ☐ A Amazon EBS with provisioned IOPS | ☐ B EC2 instance store |
| ☐ C Amazon S3 | ☐ D Amazon EFS |

◆ PAUSE AND DECIDE

Which storage should they use for these temporary files?

- A

 Amazon EBS with provisioned IOPS
Fast and persistent, but you pay for durability and persistence the team explicitly does not need.
- B

 EC2 instance store
Physically attached to the host for maximum throughput, included in the instance price, and losing it is acceptable here.
- C

 Amazon S3
Object storage over the network. Wrong shape entirely for scratch files during a computation.
- D

 Amazon EFS
Shared network filesystem. Slower than local disk and unnecessary, since nothing else reads these files.

THE SIGNAL

Willing to lose the data is the instance store signal. It is the one case where ephemeral is a feature.

A company has a large archive of research data in Amazon S3. Some datasets are downloaded weekly, others have not been touched in two years, and analysts occasionally revive an old dataset without warning.

The company wants to reduce cost but cannot predict which data will be needed, and does not want retrieval delays or fees when analysts revive something.

Which storage class should the company use?

- ☐ A S3 Standard-IA
- ☐ B S3 Glacier Flexible Retrieval
- ☐ C S3 Intelligent-Tiering
- ☐ D A lifecycle policy to Glacier Deep Archive after ninety days

◆ PAUSE AND DECIDE

Which storage class should the company use?

- A

 S3 Standard-IA
Cheaper storage, but it charges a retrieval fee, which the company said it does not want.
- C

 S3 Intelligent-Tiering
Moves objects by observed access, with no retrieval fees between tiers. Built for exactly an unpredictable pattern.
- B

 S3 Glacier Flexible Retrieval
Retrieval takes minutes to hours. The company ruled out retrieval delays.
- D

 A lifecycle policy to Glacier Deep Archive after ninety days
Age-based rules need a predictable pattern. Reviving a two-year-old dataset would take up to 48 hours.

THE SIGNAL

Cannot predict, no delays, no retrieval fees. Three constraints that only Intelligent-Tiering satisfies together.

A company still backs up its on-premises servers to physical tape, using backup software that writes to a tape library. It wants to stop handling tapes and store the backups in AWS instead.

It does not want to replace or reconfigure its existing backup software.

Which service should the company use?

- | | |
|--|--|
| ☐ A AWS Backup | ☐ B AWS Storage Gateway, Tape Gateway |
| ☐ C Amazon S3 Glacier Deep Archive directly | ☐ D AWS Elastic Disaster Recovery |

◆ PAUSE AND DECIDE

Which service should the company use?

- A

AWS Backup
Centralises backup of AWS resources. It does not present a tape interface to on-premises software.
- B

AWS Storage Gateway, Tape Gateway
Presents a virtual tape library the existing software already understands, backed by S3 and Glacier classes.
- C

Amazon S3 Glacier Deep Archive directly
The right destination, but the backup software cannot write to it without being reconfigured.
- D

AWS Elastic Disaster Recovery
Replicates servers for failover. This is about backup retention, not standing up a recovery environment.

THE SIGNAL

Does not want to reconfigure the existing software is the Storage Gateway signal in every one of its forms.

An EC2 instance in one Availability Zone has an EBS volume holding an application's data. The company is redesigning for high availability and needs an identical volume attached to a new instance in a different Availability Zone. The data must be copied across, and downtime should be minimal.

How should the company do this?

- ☐ A Detach the volume and attach it to the new instance
- ☐ B Take a snapshot and create a volume from it in the other AZ
- ☐ C Enable Multi-AZ on the EBS volume
- ☐ D Copy the data to instance store on the new instance

◆ PAUSE AND DECIDE

How should the company do this?

- A

 Detach the volume and attach it to the new instance
A volume can only attach within its own Availability Zone. Detaching does not let it cross a zone boundary.
- B

 Take a snapshot and create a volume from it in the other AZ
Snapshots live in S3 and are not zone-bound, so a new volume can be created in any AZ from one.
- C

 Enable Multi-AZ on the EBS volume
Invented. Multi-AZ is an RDS feature. EBS volumes have no such setting.
- D

 Copy the data to instance store on the new instance
Instance store is ephemeral. Application data would be lost on the next stop.

THE SIGNAL

EBS volumes cannot cross an Availability Zone. Snapshots are the only route, because they live in S3.

What to carry out of this module

- ◆ **Object, block or file first.** Cannot be modified plus mounted filesystem always eliminates S3.
- ◆ **Two facts pick an S3 class:** how often it is read, and how fast it must come back.
- ◆ Unpredictable access means Intelligent-Tiering. A known age rule means a lifecycle policy.
- ◆ EBS persists and is zone-bound. Instance store is fast and disappears. Snapshots cross zones and Regions.
- ◆ **NFS and Linux means EFS. SMB and Windows means FSx.** The protocol names the service.

ON THE EXAM

Durability is whether the data still exists. Availability is whether you can reach it. One Zone classes trade the second.